

Spatial Models, Electoral Systems, and Representational Outcomes

Marris Sobotka

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Abstract

This paper develops a computational spatial voting model to compare how First-Past-the-Post, Approval Voting, and Instant Runoff Voting shape representational outcomes in a dynamic two-dimensional electorate. Voters and parties are modeled as points in ideological space, with parties adapting their positions over repeated elections. Across 1,000 simulation runs, Approval Voting produces the lowest voter-winner distance, highest aggregate welfare, and highest Condorcet winner selection rate. The results suggest that electoral systems preserving more information about voter preferences produce outcomes that are both more representative and higher in welfare. The model remains simplified, especially in its treatment of party movement, but it shows how electoral rules can shape representation even when voter preferences are held constant.

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1 Introduction and Research Question

Electoral systems form the basis of representative democracy by translating citizen preferences into political representation. The rules that define an electoral system determine not only how winners are selected, but also how parties organize, how voters behave, and how closely election outcomes reflect the preferences of the electorate. Even small changes in electoral design can produce substantially different political outcomes. For example, plurality systems tend to produce two-party competition, while alternative systems often support a greater number of viable parties and more diverse patterns of representation. These institutional differences shape strategic behavior and influence how effectively democratic systems aggregate voter preferences.

Political scientists have long studied these dynamics using spatial models of voting. Foundational work by Black (1948) and Downs (1957) models elections as competition in an ideological space, where voters and parties are represented by points and preferences depend on distance. In the one-dimensional case under majority rule, these models produce the median voter result, in which the candidate closest to the median voter is guaranteed to win. However, real electorates are more complex. Voters hold preferences across multiple issue dimensions, and parties adapt strategically over time. When spatial models are extended to multiple dimensions, majority preferences can become unstable, and no single alternative is guaranteed to defeat all others in pairwise comparison. This instability helps explain why different electoral rules can produce different outcomes from the same underlying electorate.

Despite these theoretical insights, many models of electoral competition rely on simplifying assumptions, such as fixed party positions or static voter behavior, that limit their ability to capture the dynamics of real political systems. In particular, less attention has been given to how electoral rules interact with endogenous party movement and competition over time in multidimensional settings.

This paper addresses this gap by developing a computational model of electoral competition in a two-dimensional policy space in which parties adapt their positions over repeated elections. Using Monte Carlo simulation, I compare the performance of three electoral rules, First-Past-the-Post, Approval Voting, and Instant Runoff Voting, under identical voter distributions and dynamic party behavior. The model incorporates voter turnout, party movement, and party merging, allowing the party system to evolve endogenously under each rule. The contribution of this paper is not to claim that one electoral system is universally best, but to show how different rules preserve or discard voter preference information in a dynamic spatial model. By holding the voter distribution constant and allowing parties to adapt under different rules, the simulation isolates the effect of electoral institutions on representational outcomes.

The central research question is: how do different electoral systems shape the long-term evolution of party systems and the representativeness of election outcomes when voters have multidimensional preferences and parties adapt over time? To address this question, I evaluate each system using measures of voter–winner distance, Condorcet efficiency, and overall representational quality, and compare simulated outcomes to theoretical expectations.

2 Literature Review

This literature review examines how spatial models of voting and electoral system design shape representational outcomes in democratic systems. Political scientists have used spatial models to analyze how voter preferences, party behavior, and electoral rules interact to produce patterns of stability and instability. Understanding these dynamics is important for explaining why democratic representation varies across electoral systems and how different electoral designs translate individual preferences into political outcomes. Spatial models provide a clear theoretical framework for analyzing these differences by representing voters and parties within an ideological space, allowing scholars to study equilibrium, strategic behavior, and institutional effects. This review synthesizes research on spatial voting theory, party competition, and comparative metrics to evaluate electoral systems within a coherent theoretical framework.

2.1 Foundations of Spatial Voting Theory

Early work on electoral systems began with the idea that voter preferences can be represented along a single dimension. Hotelling (1929) first introduced this spatial idea in an economic context, illustrating that two entities attempting to maximize market share will converge toward the midpoint of consumer opinion.

Downs (1957) translated this logic to democratic politics, showing that under a plurality system, a candidate maximizing vote share should position themselves at the median voter.

Black (1948) formalized these theories mathematically by proving that when voters have single-peaked preferences, majority rule reaches an equilibrium at the median voter. Black introduced an all-pairs voting procedure in which every possible pair of alternatives is compared, showing that with an odd number of voters, the median alternative defeats all others. These results established the median voter theorem as the foundational equilibrium prediction for one-dimensional spatial models of elections.

The certainty of equilibrium in one-dimensional elections does not extend to multidimensional settings. Plott (1967) demonstrated that with more than one issue dimension, there is no median point that produces a stable equilibrium. McKelvey (1979) further showed the chaotic nature of multidimensional models, demonstrating that for any configuration of voters, an agenda setter can construct a sequence of pairwise votes that makes any point in the policy space the winner.

To represent the direction and strength of voter preferences, Enelow and Hinich (1990) and Ordeshook (1978) developed the Weighted Euclidean Distance (WED) model. In this model, voter utility is given by:

$$U = -(p - v)^T A (p - v),$$

where v is the voter’s ideal point, p is the candidate’s position, and A is a salience matrix whose diagonal entries capture issue importance and whose off-diagonal entries allow for cross-issue interactions. These developments illustrate that the median voter theorem does not extend to multidimensional models, motivating further research into conditions under which stability can emerge.

To reintroduce stability into multidimensional spatial models, scholars have examined constraints that reduce the effective decision space available to political actors. Aldrich (1983) extends the Downsian framework by shifting focus from voter choice to participation decisions. Asay (2008) formalizes the concept of a constrained strong point, a policy position that minimizes the win-set under binding ideological or organizational limits. Under such conditions, a constrained core can exist even when the unconstrained majority-rule core is empty. Hinich and Munger (1994, 1997) similarly argue that ideology functions as a constraint on party movement, generating equilibrium outcomes even in multidimensional settings. Wang, Cervas, and Grofman (2021) extend this logic through a complex-systems framework, showing how feedback between polarization, party control, and institutional design can produce persistent and path-dependent equilibria. While multidimensional majority rule is unstable in theory, stability can re-emerge when political competition is structured by institutional and ideological constraints.

2.2 Adaptive Parties and Strategic Coordination

Research on party behavior demonstrates that electoral systems strongly influence patterns of party competition. Duverger (1954) argues that plurality systems tend to produce two-party equilibria, while proportional representation encourages multiparty systems. Cox (1997, 2007) formalizes this relationship by showing that at most $M + 1$ viable parties can exist in an M -seat district, driven by strategic voting and candidate entry.

Empirical work supports these theoretical predictions. Favardin and Lepelly (2006) find evidence consistent with the $M + 1$ bound across multi-seat elections, while Kollman, Miller, and Page (1992) show through simulation that party movement and entry dynamics can produce similar equilibria.

Party entry can also be understood as a cost-benefit calculation. Feddersen, Sened, and Wright (1977) model entry as dependent on expected benefits relative to costs, while Black (1958) highlights how internal party structures can limit entry even when costs are low. Wang et al. (2021) argue that institutional feedback, including gerrymandering and incumbency advantage, further raises barriers to entry. Laver and Schofield (1990) show that strategic pressures tend to push parties toward centrist or coalition-relevant positions.

Evidence from candidate-level competition supports the spatial logic of party positioning. Brockman and Kalla (2024) demonstrate that voters respond to ideological differences among candidates, with moderate candidates often outperforming more extreme alternatives due to proximity to pivotal voters. These findings reinforce the importance of ideological distance in shaping electoral outcomes.

In addition to spatial models, axiom-based approaches highlight inherent tradeoffs in voting systems. Arrow’s impossibility theorem (1963) shows that no ranked voting system can satisfy a set of fairness criteria simultaneously. May (1952) demonstrates that majority rule uniquely satisfies key fairness conditions in two-candidate elections. Black (1976) critiques Arrow by arguing that real-world preferences are structured rather

than arbitrary. Together, these results show that voting systems involve unavoidable tradeoffs even before considering spatial complexity.

2.3 Metrics for Comparing Electoral Systems

Comparative research evaluates electoral systems using a range of metrics. Lijphart (1994) identifies electoral formula, district magnitude, and electoral thresholds as key determinants of system outcomes. Taagepera and Shugart (1989) argue that district magnitude is the primary driver of proportionality and introduce the concept of an effective threshold.

To measure party system size, Laakso and Taagepera (1979) define the effective number of parties:

$$N = \frac{1}{\sum v_i^2},$$

where v_i represents party vote shares. This measure accounts for both the number and relative size of parties.

Classification of voting systems also depends on whether they satisfy the Condorcet criterion. Black (1958) and Tideman and Plassmann (2014) distinguish between Condorcet-consistent and non-consistent rules. Their simulations show that non-consistent systems can frequently fail to select the majority-preferred candidate.

Green-Armytage, Tideman, and Cosman (2015) extend this analysis by introducing Resistance to Strategy (R-score) and Utilitarian Efficiency (E-score) as metrics for evaluating voting systems. These measures capture how systems respond to strategic behavior and how well they maximize aggregate voter utility.

Together, this literature demonstrates that representational outcomes depend on the interaction between voter preferences, party strategy, and institutional rules. Spatial voting theory explains how preference structure shapes equilibrium, while research on party coordination shows how electoral systems influence competition. Comparative metrics provide tools for evaluating how effectively electoral systems translate preferences into outcomes.

2.4 Hypotheses

Based on this literature, I test four expectations. First, FPTP should produce less representative outcomes because it records only each voter’s top choice and is vulnerable to vote splitting. Second, Approval Voting should produce more representative outcomes because it allows voters to express support for multiple acceptable parties. Third, IRV should perform between FPTP and Approval Voting because it preserves ranked information but uses that information only through sequential elimination. Fourth, because multidimensional preferences lack a guaranteed stable majority-rule equilibrium, party systems should show some variation over time, although the degree of instability may differ by electoral rule.

3 Methods

I simulate electoral competition using a spatial voting model implemented in Python, developed in Visual Studio Code. **Electoral competition** refers to the process by which political parties or candidates choose policy positions in order to attract voter support and win elections. The implementation relies on NumPy for vectorized linear algebra operations and random sampling, and Matplotlib for visualizing voter distributions, party trajectories, and winner regions. Voters and parties are represented as points in a two-dimensional ideological space, where Euclidean distance captures ideological similarity. This structure follows the spatial theory of voting developed by Black (1948), Downs (1957), and Enelow and Hinich (1984), and aligns with the empirical result of Tideman and Plassmann (2014) that spatial models closely reproduce the distributional properties of real-world ballot data.

The simulation is organized into modular files: `voters.py` generates voter ideal points; `parties.py` handles party initialization, movement, and merging; `utilities.py` computes utilities, turnout, and ballots; `metrics.py` evaluates winners and representational distances; and `main.py` runs the full iteration loop. This section describes the modeling assumptions, update rules, and performance metrics.

3.1 Voter Distribution

The electorate is generated using a two-dimensional spatial model following Tideman and Plassmann (2014). Voters are represented by ideal points in $\mathbb{R}^2$, and ideological polarization is modeled through a mixture of two circular multivariate normal distributions. A multivariate distribution describes how values are distributed across multiple variables; here, it specifies how voter preferences are jointly distributed across both ideological dimensions.

Let $\mathbf{x}_i = (x_{i1}, x_{i2})$ denote the ideal point of voter i . Each voter is first assigned to one of two ideological clusters via a Bernoulli draw: $g_i \sim \text{Bernoulli}(\pi)$, where π is the probability of belonging to the left cluster. A *Bernoulli draw* is a binary random process that assigns each voter to one of two categories. *Clusters* refer to groups of voters concentrated around similar ideological positions.

Based on this assignment, voter i 's position is drawn from one of two spherical Gaussian distributions:

$$x_i \sim \begin{cases} \mathcal{N}(\mu_L, \sigma_L^2 I_2), & g_i = 1, \\ \mathcal{N}(\mu_R, \sigma_R^2 I_2), & g_i = 0, \end{cases}$$

where the cluster centers are fixed at $\mu_L = (-1.5, -0.5)$, $\mu_R = (1.5, 0.5)$. A *Gaussian (normal) distribution* is a symmetric probability distribution centered at a mean, where values closer to the center are more likely. Each component distribution is unimodal, meaning it has a single peak, and symmetric around its center.

This mixture formulation produces a bimodal electorate and aligns with observations that ideological distributions often exhibit clustering rather than being unimodal or symmetric. A bimodal distribution has two distinct peaks, reflecting the presence of two groups of voters. The simulation also supports a single-cluster electorate by drawing all voters from a single circular Gaussian distribution when polarization is not desired.

3.2 Party Initialization

Parties are represented as points in the same two-dimensional policy space as voters. Let $\theta_p = (\theta_{p1}, \theta_{p2})$

denote the ideological position of party p . At the beginning of the simulation, each party is placed by drawing from a spherical normal distribution centered at the origin and σ_θ is the spread parameter:

$$\theta_p \sim \mathcal{N}((0, 0), \sigma_\theta^2 I_2).$$

This initialization is a simplifying assumption used to isolate the effects of voter distributions and electoral rules. By starting all parties near the center of the policy space, any divergence, clustering, or persistence in party positions emerges endogenously from voter preferences and the dynamics of the model, rather than from initial conditions.

3.3 Utility Model

Voter utilities follow a multidimensional spatial formulation in which utility decreases with ideological distance. Voter utility refers to the level of satisfaction a voter derives from a party's position, with higher values indicating stronger preference. Let x_i denote the ideal point of voter i and θ_p the position of party p . Ideological distance is the Euclidean distance between a voter's ideal point and a party's position, representing the degree of disagreement in the policy space. The baseline component of utility is negative squared Euclidean distance:

$$U_i^{\text{base}}(p) = -\|x_i - \theta_p\|^2.$$

This widely used formulation dates to Black (1948) and Downs (1957) and is the most used in multidimensional spatial models (Enelow and Hinich 1984; Tideman and Plassmann 2014).

To create interactions between issue dimensions, I incorporate a non-separable component following Stoetzer and Zittlau (2015). In their framework, preferences over one policy dimension may depend on a voter's position on another dimension, leading to cross-dimensional interaction effects. The full utility function in the simulation is therefore

$$U_i(p) = -[(x_{i1} - \theta_{p1})^2 + (x_{i2} - \theta_{p2})^2 + \gamma(x_{i1} - \theta_{p1})(x_{i2} - \theta_{p2})].$$

where γ governs the strength and direction of the interaction.

I set $\gamma = 0.5$ in all simulations reported here, introducing moderate non-separability. With $\gamma > 0$, voters particularly dislike parties that differ from them in the *same* direction on both dimensions simultaneously (e.g., a voter who is left/libertarian perceives a right/authoritarian party as especially distant). This interaction term bends indifference curves away from circles and affects which parties voters view as close substitutes. Activating non-separable preferences also changes party learning dynamics, turnout patterns, and the likelihood of Condorcet cycles, which is consistent with the findings of Stoetzer and Zittlau (2015) that multidimensional interactions increase the complexity of majority preferences.

3.4 Ballot Generation Under Three Voting Systems

Given utilities $U_i(p)$ from the spatial model, each voter i produces a ballot according to the rule of the electoral system. An *electoral rule* is the procedure used to translate individual voter preferences into a collective outcome. I implement three common rules, First-Past-the-Post, Approval Voting, and Instant Runoff Voting, which differ in how they allow voters to express preferences and how those preferences are aggregated. Let $\mathcal{P}$ denote the set of parties and $P = |\mathcal{P}|$.

First-Past-the-Post (FPTP)

First-Past-the-Post (FPTP) is a plurality voting system in which each voter selects a single candidate, and the candidate with the most votes wins. Under plurality rule, each voter selects the party that maximizes utility:

$$\text{ballot}_i^{\text{FPTP}} = \arg \max_{p \in \mathcal{P}} U_i(p).$$

Because utility decreases monotonically with distance, meaning that greater ideological distance always corresponds to lower utility, this is equivalent to selecting the nearest party in the ideological space.

Approval Voting

Approval Voting allows voters to support (approve of) any number of candidates, and the candidate with the most approvals wins. For Approval Voting, voters may support multiple parties. Rather than imposing a global utility cutoff, I use a voter-specific quantile threshold following Laslier’s (2006) spatial approval framework. For voter i , the approval threshold is the q -quantile of their utility vector:

$$T_i = \text{quantile}_q(\{U_i(p) : p \in \mathcal{P}\}).$$

with $q = 0.5$ in the baseline simulations. Setting $q = 0.5$ means that each voter approves the top half of parties according to their utility ranking, subject to ties at the threshold. This provides a neutral benchmark that avoids imposing an arbitrary fixed cutoff and ensures that approval behavior depends only on relative preferences.

Binary approval decisions are then given by

$$\text{Approve}_i(p) = \begin{cases} 1, & U_i(p) \geq T_i, \\ 0, & U_i(p) < T_i, \end{cases}$$

yielding a P -dimensional approval vector for each voter.

Instant Runoff Voting (IRV)

Instant Runoff Voting (IRV) is a ranked-choice voting system in which voters rank candidates, and the winner is determined through a sequence of eliminations and vote transfers until one candidate achieves a majority. IRV requires each voter to submit a complete ranking of parties. Because utilities produce a strict ordering, rankings are generated by sorting parties in descending utility order:

$$p_1 \succ_i p_2 \succ_i \dots \succ_i p_P \iff U_i(p_1) > U_i(p_2) > \dots > U_i(p_P).$$

The corresponding ballot is implemented as:

$$\text{ballot}_i^{\text{IRV}} = \text{argsort}(-U_i).$$

which returns a permutation of all P parties sorted by voter i 's preferences.

During IRV tabulation, first-choice votes are counted, the party with the fewest first-choice votes is eliminated, and ballots are transferred to each voter's next preferred active party. This process repeats until one party obtains a majority of active ballots. Ties in last place are resolved deterministically using array order; the implementation does not employ randomized tie-breaking.

3.5 Party Movement

After each simulated election, parties update their positions toward the mean location of the voters who supported them. Let $S_p = \{i : p = \arg \max_{q \in \mathcal{P}} U_i(q)\}$ denote the set of supporters of party p . The centroid of these supporters is

$$\bar{x}_p = \frac{1}{|S_p|} \sum_{i \in S_p} x_i.$$

Parties adjust their positions via a simple learning rule with step size η : $\theta_p^{(t+1)} = \theta_p^{(t)} + \eta (\bar{x}_p^{(t)} - \theta_p^{(t)})$. If a party receives no supporters in iteration t , its position remains unchanged, because there is no information about voter preferences to guide adjustment.

This update rule represents a deterministic vote-seeking adjustment: parties move toward the center of their current support base. Each new party position is determined by the party's previous position, the current assignment of voters to parties, and the fixed voter distribution. In this sense, the model has a Markov-like structure because the next configuration of party positions depends on the current configuration rather than the full history of earlier iterations. However, because the simulation uses a deterministic update rule rather than estimating transition probabilities, I treat the process as an adaptive spatial dynamic rather than a fully specified Markov process. This approach of repeatedly adjusting positions in space reflects the general logic of spatial models, which examine party behavior and system stability by tracking how they move toward equilibrium configurations (Aldrich, 1983).

3.6 Party Merging

After party positions are updated, parties that occupy nearly identical ideological locations are merged. Let the Euclidean distance between two parties p and q be $d(p, q) = \|\theta_p - \theta_q\|$. If this distance falls below a merging threshold d_{merge} , the parties are treated as members of the same ideological cluster:

$$d(p, q) < d_{\text{merge}} \implies p, q \in C.$$

For any such cluster C of near-duplicate parties, the merged party replaces its members and is positioned at their centroid:

$$\theta_{\text{merged}} = \frac{1}{|C|} \sum_{p \in C} \theta_p. \quad (1)$$

This rule reflects a core implication of Euclidean spatial voting theory: parties that are extremely close in ideological space are indistinguishable to voters (Downs 1957; Cox 1997). They attract the same supporters, generate the same utilities, and produce identical ballots under all voting systems. Allowing such parties to persist separately would artificially inflate the number of political alternatives without offering meaningful policy differentiation.

The implemented algorithm uses a greedy clustering procedure. In this context, "greedy" means that the algorithm builds clusters step by step using only local information, without reconsidering earlier grouping decisions. Iterating through parties in order, each unvisited party initiates a cluster consisting of all other parties within distance d_{merge} . Once a party is assigned to a cluster, it is not reconsidered for inclusion in another cluster, even if a different grouping might produce a better overall partition.

Each cluster is replaced by a single merged party located at the centroid given by Equation (1). Because each party is visited once, this procedure produces at most one merged party per cluster per iteration.

This approach is computationally efficient and ensures that nearby parties are consolidated, but it does not guarantee that the resulting clustering is globally optimal.

Merging is applied once per simulation iteration after party updates. After merged positions are computed, no further merging occurs until the next election cycle.

3.7 Turnout

To incorporate abstention into the simulation, I model turnout as a decreasing function of ideological distance. Let $d_i = \min_{p \in \mathcal{P}} \|x_i - \theta_p\|$ denote the distance between voter i and the nearest party. Following spatial models of political participation (Aldrich 1983), voters are assumed to be less likely to vote when no party is meaningfully close to their preferences.

I convert this distance into a turnout probability using an exponential decay function:

$$\Pr(\text{vote}_i = 1) = \exp(-\alpha d_i). \quad (2)$$

where $\alpha > 0$ determines how sharply turnout falls with ideological distance. This formulation ensures that turnout is highest among voters located near a party and declines rapidly for voters who are far from all parties.

For each voter, a Bernoulli draw using the probability in Equation (2) determines whether the voter participates in the election. Because party positions change each iteration, turnout is recalculated at every time step to reflect voters' changing proximity to the evolving party system.

All election outcomes (FPTP, Approval, IRV) are computed using only voters for whom $\text{vote}_i = 1$.

To assess representativeness, I track the mean ideological distance between each winning party w and the voters who actually turned out:

$$R(w) = \frac{1}{N_{\text{voted}}} \sum_{i: \text{vote}_i = 1} \|x_i - \theta_w\|.$$

This measure summarizes how close the winning party is to participating voters in each iteration.

3.8 Outcome Metrics

At each iteration t of the simulation, I record a set of statistics that describe both the structure of the party system and the performance of each voting rule. These metrics are derived from the spatial configuration of voters and parties and correspond directly to the outputs computed in `metrics.py`. Specifically, I track:

- the number of active parties P_t after merging,
- the movement of parties between iterations,
- the number of party merges in iteration t ,
- the mean distance between voters and their nearest party,
- the mean distance between each party and its supporters,
- the winning party under FPTP, Approval Voting, and IRV,
- whether election outcomes result in ties under each rule,
- the existence of a Condorcet winner and whether each rule selects it.

Mean distances are computed using Euclidean distance in the policy space. The distance to the nearest party captures overall representational coverage, while supporter distance measures how well each party represents its own base.

A *Condorcet winner* is a candidate who would defeat every other candidate in pairwise majority comparisons. Condorcet winners are computed directly from the utility rankings by comparing each pair of parties across voters. Following Tideman and Plassmann (2014), the frequency with which a voting rule selects the Condorcet winner is used as one benchmark of rule quality.

For each electoral rule, I also record the winning party, whether the outcome involves a tie, and the mean ideological distance between the winner and voters. These rule-specific outputs allow for direct comparison of how different systems translate voter preferences into outcomes.

Party movements, merges, and distance measures allow the simulation to illustrate how party systems evolve over time and how representativeness shifts across iterations. These recorded statistics support visualizations such as party trajectories, winner-region diagrams, and distributions of voter–winner distances, enabling systematic comparison of electoral-rule performance in a dynamic spatial environment.

3.9 Parameter Choices

The simulation requires selecting several parameters that shape the behavior of voters and parties. These choices are grounded in both theoretical considerations and practical goals for the model.

Number of voters. I use $N = 1000$ voters. This provides a sufficiently dense approximation of the spatial distribution while maintaining computational feasibility across repeated simulation runs.

Initial number of parties. The model begins with $P_{\text{init}} = 4$ parties. This reflects a multiparty setting that allows for differentiation, movement, and merging, while remaining small enough for trajectories to be interpretable. Starting with multiple parties allows the number of viable competitors to emerge endogenously under different electoral rules.

Learning rate for party movement. Parties update their positions using a learning rate of $\eta = 0.3$. Values near zero produce slow convergence, while values near one cause large jumps toward voter centroids. A moderate value ensures gradual but responsive adjustment.

Merging threshold. The merging threshold is set to $d_{\text{merge}} = 0.5$. This defines the minimum ideological separation required for parties to remain distinct. Parties within this distance are treated as effectively identical and are merged.

Turnout sensitivity parameter. Turnout is determined by $P(\text{vote}_i = 1) = \exp(-\alpha d_i)$ with $\alpha = 1.0$. Larger values of α generate steeper declines in turnout as ideological distance increases, while smaller values produce more uniform participation. This value produces meaningful variation without eliminating participation from distant voters.

Non-separable preference parameter. The spatial interaction term uses $\gamma = 0.5$ to incorporate cross-dimensional dependence in preferences. Positive values introduce interaction effects across dimensions, altering the shape of voter indifference curves and increasing the complexity of majority relations.

Number of iterations. Each simulation is run for $T = 100$ iterations, allowing party systems sufficient time to evolve through repeated cycles of competition, movement, and merging.

Number of simulation runs. To ensure that results are not driven by random variation in voter draws or initial conditions, I run the simulation 1000 times with different random seeds. Aggregate statistics are computed across these runs, including mean outcomes, standard deviations, and confidence intervals. This Monte Carlo approach allows for robust comparison of electoral rules across a wide range of simulated electorates.

4 Results

This section presents the results of the simulation analysis, comparing the performance of First-Past-the-Post (FPTP), Approval Voting, and Instant Runoff Voting (IRV) across measures of representation, welfare, and rule consistency. All results are based on 1,000 simulation runs of 100 iterations each.

4.1 System-Level Dynamics

The simulation produces stable party systems across runs. The average number of parties at the end of each simulation is approximately 3.962 (SD = 0.191), with minimal variation across iterations. Party merging is rare, averaging 0.038 merges per simulation run, and party movement is limited, with an average movement of 0.017 units per iteration.

This result partially contradicts the expectation that multidimensional competition would produce instability. In this simulation, instability appears more strongly in rule choice and winner selection than in the number or location of parties. This likely reflects the simplifying movement rule, which pulls parties toward supporter centroids and encourages convergence once stable support bases form.

Figure 1 illustrates the voter distribution and final party positions in a representative simulation run. Voters form two main ideological clusters, and parties settle in positions that reflect these underlying concentrations.

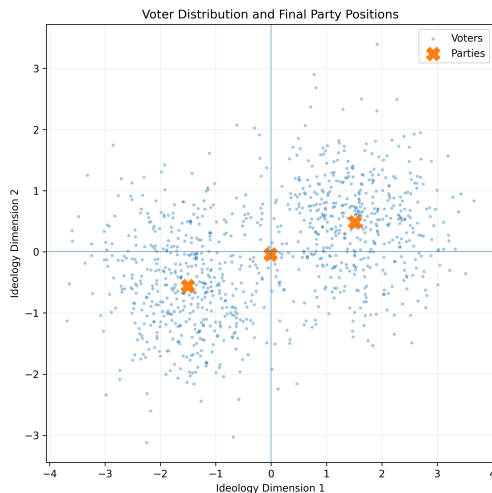


Figure 1: Voter distribution and final party positions in a representative simulation run.

Figure 2 shows party movement over time under Approval Voting. Parties initially move toward voter clusters and then stabilize, consistent with the low average movement observed across simulations.

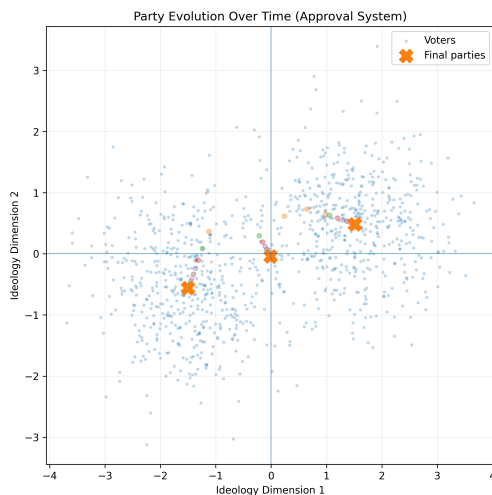


Figure 2: Party evolution over time under Approval Voting.

4.2 Divergence Across Electoral Rules

Despite identical voter distributions, electoral rules frequently produce different outcomes. Agreement rates between rules are moderate: FPTP and Approval agree in 40.7% of elections, FPTP and IRV in 54.1%, and Approval and IRV in 57.1%.

These results show that electoral rules systematically influence which party wins, even when underlying preferences are held constant. Within the simulation, institutional design plays a central role in determining which party wins from the same underlying voter distribution.

4.3 Representational Performance

Representational quality is measured as the mean ideological distance between voters and the winning party, where lower values indicate better representation.

FPTP produces the largest voter–winner distance (Mean = 2.154, SD = 0.228), followed by IRV (Mean = 2.021, SD = 0.146), while Approval Voting produces the smallest distance (Mean = 1.944, SD = 0.062).

Figure 3 shows the distribution of final voter–winner distances across simulation runs. Approval Voting not only has the lowest mean distance but also the least variation, while FPTP exhibits both higher distances and greater dispersion.

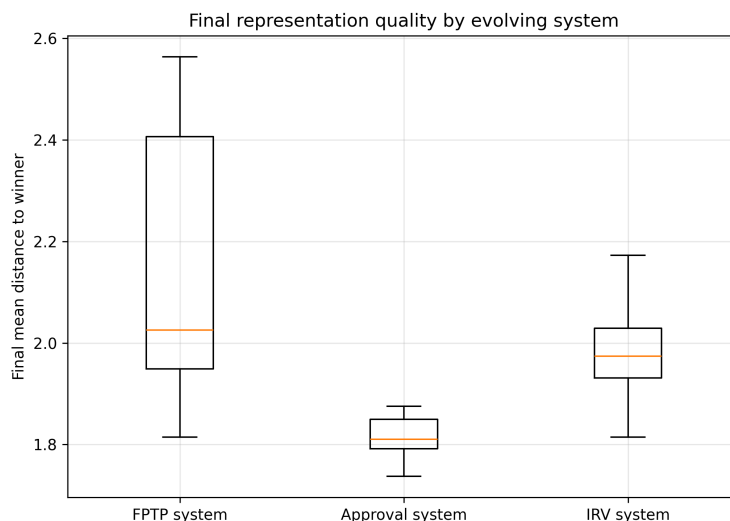


Figure 3: Final voter–winner distance by electoral rule. Lower values indicate better representation.

All pairwise differences are highly statistically significant. Using mean voter–winner distance, FPTP is significantly farther from voters than Approval Voting ($t = 27.993$, $p < 0.001$) and IRV ($t = 15.544$, $p < 0.001$). Approval Voting is also significantly closer to voters than IRV ($t = -15.176$, $p < 0.001$, comparing Approval minus IRV).

These results indicate that Approval Voting consistently produces more representative outcomes, while FPTP performs worst on this measure.

4.4 Welfare Outcomes

Aggregate welfare is measured as the average utility voters derive from the winning party. Because utility is based on negative squared distance, values closer to zero indicate better outcomes.

Approval Voting produces the highest welfare (Mean = -5.285), followed by IRV (Mean = -5.889), and FPTP produces the lowest welfare (Mean = -7.070). Approval Voting also selects the welfare-maximizing candidate in 63.3% of cases.

Figure 4 shows the distribution of welfare outcomes. Approval Voting produces both higher and more consistent welfare than the other rules.

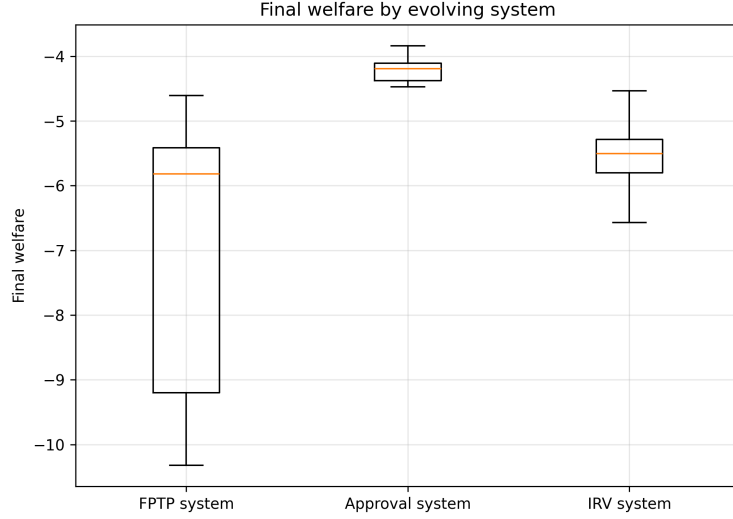


Figure 4: Final welfare by electoral rule. Values closer to zero indicate better outcomes.

All pairwise comparisons are statistically significant. FPTP performs significantly worse than Approval Voting ($t = -29.473$, $p < 0.001$) and IRV ($t = -16.974$, $p < 0.001$), and Approval Voting performs significantly better than IRV ($t = 15.649$, $p < 0.001$).

These results show that Approval Voting produces the highest average welfare among the three rules.

4.5 Condorcet Efficiency

The frequency with which each rule selects the Condorcet winner also varies substantially. When a Condorcet winner exists, Approval Voting selects it in 74.7% of cases, compared to 47.6% for IRV and 34.6% for FPTP.

Figure 5 shows this pattern directly, with Approval Voting achieving the highest Condorcet efficiency.

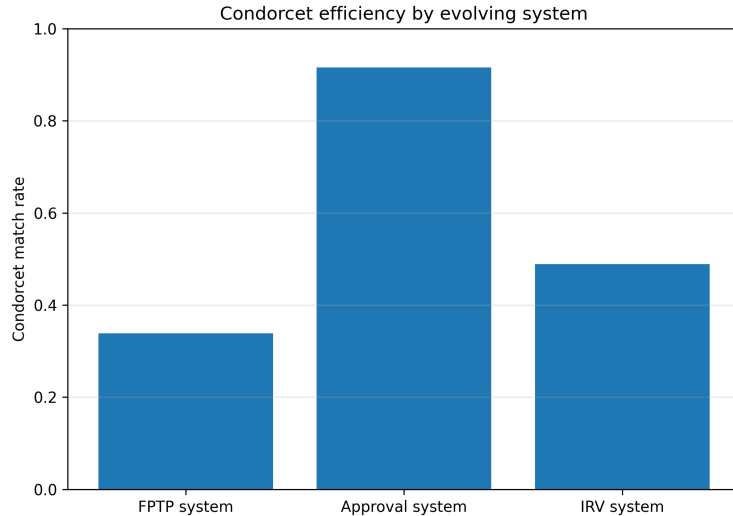


Figure 5: Condorcet winner selection rate by electoral rule.

This indicates that Approval Voting is more likely to select candidates that are majority-preferred in pairwise comparisons, while FPTP does so least often. However, selecting the Condorcet winner does not always correspond to maximizing welfare, highlighting a potential tension between majority consistency and

aggregate utility.

4.6 Representation and Welfare

A central result of the simulation is the alignment between representational quality and welfare. Electoral rules that minimize voter–winner distance also produce higher welfare outcomes. Approval Voting performs best on both dimensions, while FPTP performs worst, with IRV consistently in between.

Together, Figures 3 and 4 show that more representative outcomes are also more efficient within this model, with no observed tradeoff between representation and aggregate welfare under these conditions.

4.7 Summary of Findings

Across the main representational, welfare, and Condorcet metrics, Approval Voting consistently outperforms both IRV and FPTP. It produces more representative outcomes, higher welfare, and more frequent selection of Condorcet winners. IRV improves on FPTP but remains less effective than Approval Voting across these measures. FPTP performs worst, producing less representative and lower-welfare outcomes.

These findings support the hypothesis that electoral rules allowing voters to express broader preferences produce outcomes that better reflect the underlying distribution of voter preferences.

5 Discussion

The results of the simulation highlight a consistent pattern across electoral rules: systems that keep more information about voter preferences produce outcomes that are both more representative and higher in aggregate welfare. This result comes from how the utility function works and how each voting rule uses the information from the utility matrices.

5.1 Information Use and Electoral Rules

The underlying structure of the model depends on the utility function, which encodes the distance between voters and parties in a two-dimensional ideological space. This utility function provides a complete ordering of preferences over all parties for each voter. However, electoral rules differ in how much of this information they retain.

FPTP reduces the utility structure to a single choice, preserving only each voter’s top preference. As a result this method loses most of the utility distribution’s information. This compression leads to outcomes that are more sensitive to slight differences in top preferences and less responsive to broader voter alignment.

IRV preserves more information by incorporating ranked preferences, allowing secondary choices to influence outcomes through vote transfers. However, it only uses this information sequentially and conditionally, meaning that it does not consider all preference information simultaneously.

Approval Voting uses the most information by allowing voters to express a set of acceptable candidates. Because voters approve all candidates above a threshold, this system incorporates more information about voter preferences. The simulation results suggest that this increased information retention produces outcomes that are closer to voters and higher in aggregate welfare.

5.2 Representation, Welfare, and Information Preservation

The results show that representational quality and welfare move together in the simulation. Both measures originate from the same underlying utility structure, and both improve when electoral rules better reflect the distribution of voter preferences.

Approval Voting performs best across both metrics because it collects more information about voter utility. By allowing voters to support multiple candidates, it reduces the loss of information that occurs under other voting rules. This leads to winners that are closer to voters on average and generate higher total utility.

In contrast, FPTP consistently performs worst because it relies on the least amount of information. By limiting voters to only a single candidate, it ignores the broader structure of preferences and produces outcomes that are less representative and less efficient.

These results suggest that the relationship between representation and welfare depends heavily on how electoral systems process information. When more information is preserved, the two goals align; when information is lost, both representation and welfare suffer.

5.3 Implications for Real-World Electoral Systems

Although the simulation is abstract, the results have implications for real-world electoral design. Many existing electoral systems, particularly plurality systems like FPTP, require voters to express highly compressed preferences. This can lead to outcomes that do not fully reflect the underlying distribution of voter preferences.

The results suggest that systems allowing for more expressive ballots, such as Approval Voting or other multi-preference systems, may better capture voter intent and produce outcomes that are both more representative and more efficient. In particular, the ability to support multiple candidates appears to reduce the distortion introduced by strategic voting and vote splitting.

However, real-world electoral systems involve additional constraints not captured in the model, including institutional complexity, voter understanding, and political feasibility. As a result, while the simulation highlights theoretical advantages of more expressive systems, these advantages must be evaluated in the context of practical implementation.

5.4 Limitations and Model Assumptions

The simulation relies on several simplifying assumptions that limit the scope of the conclusions. Most importantly, party behavior is modeled using a simple adaptive rule in which parties move toward the mean position of their supporters. This approach captures basic responsiveness but does not account for strategic positioning, entry and exit of parties, campaign effects, incumbency, ballot access, or institutional constraints that shape real-world party competition.

Additionally, voter preferences are represented in a fixed two-dimensional ideological space with a specified distribution. While this allows for clear analysis, it abstracts away from the complexity of real political preferences, which may be higher-dimensional and context-dependent. The model also assumes sincere voting, a fixed approval threshold, and deterministic party movement. These assumptions are useful because they isolate the effect of electoral rules, but they also mean that the results should be interpreted as theoretical tendencies rather than direct predictions about real elections.

5.5 Conclusion

Within the limitations of the model, the results demonstrate that electoral rules differ fundamentally in how they process voter information. Systems that preserve more of the underlying utility structure produce outcomes that are both more representative and higher in aggregate welfare. This suggests that the design of electoral rules plays a critical role not only in determining winners but also in shaping how effectively voter preferences are translated into collective outcomes.

A Code Repository and Reproducibility

The full code used for this simulation is available in the public GitHub repository:

<https://github.com/mwsobotka/capstone>

The simulation is implemented in Python. The repository contains scripts for generating voters, initializing and updating parties, computing utilities, simulating election rules, recording outcome metrics, and producing figures. The main simulation is run through `main.py`, which generates voters, initializes parties, runs the election loop for FPTP, Approval Voting, and IRV, saves party histories and iteration statistics, and stores each simulation run as a `.pkl` file.

The core files are organized as follows:

- `main.py`: runs the full simulation. It generates voters, initializes parties, separates party systems by electoral rule, computes utilities and turnout, updates party positions, merges nearby parties, records metrics, and saves simulation outputs.
- `voters.py`: generates voter ideal points. The model supports both a single normal voter distribution and a two-cluster mixture distribution.
- `parties.py`: initializes party positions, updates party locations under FPTP, Approval Voting, and IRV, and merges parties that are within the specified distance threshold.
- `utilities.py`: computes spatial utilities, probabilistic turnout, FPTP ballots, approval ballots, ranked ballots, and winners under each electoral rule.
- `metrics.py`: calculates party movement, voter-party distances, supporter distances, Condorcet winners, rule-specific winners, ties, and per-iteration summary statistics.
- `analysis.py`: loads saved simulation runs and calculates aggregate summary statistics, including mean voter-winner distance, welfare, Condorcet selection rates, tie rates, party-system outcomes, rule agreement rates, and statistical tests.
- `plots.py` and `runplots.py`: generate the figures used to visualize voter distributions, party movement, voter-winner distance, welfare, Condorcet efficiency, and party-system dynamics.

The main results reported in this paper are based on 1,000 simulation runs of 100 iterations each. Each run uses a two-dimensional mixture voter distribution, four initial parties, a party learning rate of $\eta = 0.3$, a merging threshold of $d_{\text{merge}} = 0.5$, a turnout sensitivity parameter of $\alpha = 1.0$, and a non-separable preference parameter of $\gamma = 0.5$.

References

- Aldrich, J. H. (1983). A downsian spatial model with party activism. *The American Political Science Review* 77(4), 974–990.
- Arrow, K. J. (1963). *Social Choice and Individual Values* (2 ed.). New York: Wiley.
- Asay, G. R. B. (2008). How does ideology matter in the spatial model of voting? *Public Choice* 135(3/4), 109–123.
- Bergson, A. (1958). Review of an economic theory of democracy, by anthony downs. *The American Economic Review* 48(3), 437–440.
- Bhaskar, U. and S. Pyne (2024). Equilibrium computation in the hotelling-downs model of spatial competition. arXiv preprint. December 17.
- Black, D. (1948). On the rationale of group decision-making. *Journal of Political Economy* 56(1), 23–34.
- Black, D. (1998). *The Theory of Committees and Elections by Duncan Black and Committee Decisions with Complementary Valuation by Duncan Black and R. A. Newing* (2 ed.). Springer Dordrecht.
- Cox, G. W. (2007). *Making Votes Count: Strategic Coordination in the World’s Electoral Systems* (Digital printing ed.). Political Economy of Institutions and Decisions. Cambridge University Press.
- Downs, A. (1957). *An Economic Theory of Democracy*. New York: Harper.
- Duverger, M. (1969). *Political Parties: Their Organization and Activity in the Modern State* (3 ed.). University Paperbacks. Methuen.
- Enelow, J. M. and M. J. Hinich (Eds.) (1990). *Advances in the Spatial Theory of Voting*. Cambridge University Press.
- Gehrlein, W. V. and F. Valognes (2001). Condorcet efficiency: A preference for indifference. *Social Choice and Welfare* 18(1), 193–205.
- Green-Armytage, J., T. N. Tideman, and R. Cosman (2016). Statistical evaluation of voting rules. *Social Choice and Welfare* 46(1), 183–212.
- Hotelling, H. (1929). Stability in competition. *The Economic Journal* 39(153), 41–57.
- Kosar, K. R. (2023). Conservatives should look more closely at systemic election reforms. Technical report, American Enterprise Institute.
- Laslier, J.-F. (2006). Spatial approval voting. *Political Analysis* 14(2), 160–185.
- Latner, M. (2019). Fighting back: The path to democratic renewal. Technical report, Union of Concerned Scientists.
- Lijphart, A. and D. Aitkin (1994). *Electoral Systems and Party Systems: A Study of Twenty-Seven Democracies, 1945-1990*. Comparative European Politics. Oxford University Press.
- May, K. O. (1952). A set of independent necessary and sufficient conditions for simple majority decision. *Econometrica* 20(4), 680–684.
- Ornstein, J. T. and R. Z. Norman (2014). Frequency of monotonicity failure under instant runoff voting: Estimates based on a spatial model of elections. *Public Choice* 161(1/2), 1–9.
- Plassmann, F. and T. N. Tideman (2014). How frequently do different voting rules encounter voting paradoxes in three-candidate elections? *Social Choice and Welfare* 42(1), 31–75.
- Stoetzer, L. F. and S. Zittlau (2015). Multidimensional spatial voting with non-separable preferences. *Political Analysis* 23(3), 415–428.

- Taagepera, R. and M. S. Shugart (1989). *Seats and Votes: The Effects and Determinants of Electoral Systems*. Yale University Press.
- Tideman, T. N. and F. Plassmann (2011). Modeling the outcomes of vote-casting in actual elections. Technical Report 1627787, Social Science Research Network. SSRN Scholarly Paper No. 1627787, January 18.
- Tideman, T. N. and F. Plassmann (2014). Which voting rule is most likely to choose the “best” candidate? *Public Choice* 158(3/4), 331–357.
- Volie, I. (2024). The best way to choose the winner. In *Making Democracy Count*. Princeton University Press.
- Wang, S. S.-H., J. Cervas, B. Grofman, and K. Lipsitz (2021). A systems framework for remedying dysfunction in us democracy. *Proceedings of the National Academy of Sciences of the United States of America* 118(50), 1–8.